

AI PROMPT ENGINEERING

FACULTY LEARNER GUIDE & PROMPT LIBRARY

From Everyday Academic Pain Points to Reusable AI Workflows

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About this guide

This learner guide is designed as an enterprise-style learning resource for experienced university and engineering faculty. It assumes participants already know how to open and use generative-AI tools. The focus is therefore not basic AI awareness, but the capability to design, test, validate and reuse prompts for real academic work.

Learning design principle	What it means for the learner
Performance first	Start from a real faculty task or pain point rather than from a prompting technique.
Evidence first	Identify the documents, data or policy that should ground the answer.
Structured output	Specify the form in which the result must be delivered so it can be reviewed and reused.
Validation by design	Build a challenge/check step into the prompt instead of trusting fluent output.
Human accountability	AI assists analysis and drafting; faculty retain academic and institutional judgment.
Reusable capability	Convert successful task prompts into parameterized master prompts.

Learning objectives

By the end of this guide, participants should be able to:

- Explain prompt engineering as a task-design capability rather than a collection of magic phrases.
- Select an appropriate prompting technique based on task complexity, evidence requirements and output needs.
- Convert a recurring faculty pain point into an execution-ready prompt.
- Design prompts that specify role, goal, context, sources, method, constraints, output and validation.

- Use structured and evidence-grounded prompts for teaching, research, assessment and academic administration.
- Critically review AI output for unsupported claims, missing evidence, contradictions and over-generalization.
- Iterate a working prompt and derive a reusable master prompt with placeholders and quality controls.

Recommended learning path

Stage	Capability	Participant action	Evidence of learning
1	Recognize the task	Identify one recurring faculty pain point.	Pain-point statement
2	Choose the technique	Select prompting techniques appropriate to the task.	Technique selection rationale
3	Engineer the prompt	Build the prompt using the faculty prompt architecture.	Version 1 prompt
4	Execute	Run the prompt against a supplied sample file.	AI output
5	Validate	Challenge the output and identify weaknesses.	Validation notes
6	Improve	Modify instructions, evidence rules or output schema.	Version 2 prompt
7	Standardize	Convert the tested prompt into a reusable master prompt.	Master prompt

1. What is Prompt Engineering?

Prompt engineering is the deliberate design of instructions and interaction patterns that guide an AI system toward a useful, reliable and reviewable outcome. In professional learning and development terms, it is best understood as a form of task analysis: the faculty member translates an intended outcome into a clear AI-enabled workflow.

Dimension	Weak task request	Engineered task
Outcome	"Help me with my thesis."	"Identify the five highest-impact methodology weaknesses and provide

Dimension	Weak task request	Engineered task
		evidence and revision actions.”
Context	Little or no context.	Research stage, audience, discipline, purpose and constraints are specified.
Evidence	AI answers from general knowledge.	Approved files/sources are identified and treated as the evidence base.
Process	“Do everything.”	The task is decomposed into extraction, comparison, analysis and validation.
Output	Free-form prose.	A defined matrix, checklist, blueprint or action plan.
Risk control	No boundaries.	Explicit rules against fabricated citations, unsupported claims or unauthorized decisions.
Quality control	Trust the first answer.	Adversarial challenge and human validation are built into the workflow.

2. Prompting Techniques — A Faculty Decision Guide

Prompting techniques are tools, not ends in themselves. The correct technique depends on what the faculty member is trying to achieve. The table below is intentionally more descriptive than a simple technique glossary: it links each technique to the learning/work problem it solves, the conditions under which it adds value, and the limitation to watch for.

Technique	Definition	When it adds value	Faculty examples	Primary risk / limitation
Zero-shot	Give a clear task without demonstration examples.	Straightforward extraction, summarization or transformation where the desired pattern is obvious.	Summarize a meeting; extract key themes; convert notes to bullets.	May produce generic or differently formatted outputs.
Few-shot	Provide representative input-output examples before asking for the new task.	When consistency of classification, tone, format or rubric application matters.	Classify student feedback; format research-gap statements; apply a feedback rubric.	Poor examples can teach the wrong pattern.
Role / perspective	Specify the expertise or	When the quality of judgment	PhD examiner;	A role does not create real-world

Technique	Definition	When it adds value	Faculty examples	Primary risk / limitation
	viewpoint from which the task should be evaluated.	depends on perspective or evaluation criteria.	journal reviewer; curriculum designer; accreditation reviewer.	expertise or verification.
Context	Provide purpose, audience, stage, constraints and surrounding situation.	When the same task would produce different answers in different academic contexts.	Design activity for first-year vs postgraduate students; revise a proposal for a funding call.	Too much irrelevant context can dilute the task.
Grounded / source-based	Explicitly anchor the task to supplied or approved evidence.	Research, thesis review, accreditation, policy, assessment and document comparison.	Use uploaded papers only; compare a proposal with a grant call.	Grounding only works as well as the supplied/accessible sources.
Decomposition	Split a complex task into logical sub-tasks.	Large document analysis, research synthesis, curriculum redesign and multi-criterion review.	Extract → compare → identify gaps → prioritize → report.	Too many steps can make the workflow cumbersome.
Plan-and-solve	Ask AI to establish an analysis plan before executing the task.	Complex or ambiguous assignments where a visible approach improves reviewability.	Plan a literature review analysis; plan a curriculum audit.	A plan is still a hypothesis; it needs human review.
Structured output	Specify fields, columns, sections or a schema for the response.	Whenever the output will be reviewed, compared, transferred or reused.	Evidence matrix; assessment blueprint; action register.	A rigid schema can hide nuance if designed poorly.
Constraint prompting	Define boundaries, exclusions and non-negotiable rules.	Academic integrity, privacy, assessment and evidence-sensitive tasks.	Do not invent citations; do not change official CO wording.	Constraints reduce but do not eliminate AI errors.
Iterative	Use follow-up	Most meaningful	Clarify →	Users can iterate

Technique	Definition	When it adds value	Faculty examples	Primary risk / limitation
prompting	prompts to improve a result rather than expecting perfection in one turn.	academic workflows.	deepen → shorten → challenge → finalize.	endlessly without a clear quality criterion.
Adversarial validation	Ask AI to search for reasons its own answer may be wrong.	Research gaps, methodology, proposals, evidence claims and high-impact recommendations.	Try to disprove the identified research gap.	AI can miss flaws; adversarial output is not independent peer review.
Prompt chaining	Use multiple linked prompts where each stage produces input for the next.	Repeatable research and academic workflows.	Evidence extraction → synthesis → reviewer critique → executive summary.	More complex to manage and version.

3. The Recommended Faculty Prompt Architecture

For recurring faculty work, use a structured prompt as the default. Not every prompt needs every field, but the more consequential or evidence-sensitive the task, the more of these elements should be explicit.

Layer	Design question	What a strong specification looks like	Common failure if omitted
ROLE	What perspective should AI adopt?	"Act as an external PhD examiner."	Output may lack the desired evaluation lens.
GOAL	What exact result is required?	"Identify the five highest-impact weaknesses."	AI produces a broad, unfocused response.
CONTEXT	What situation, audience and stage matter?	"This is a doctoral methodology chapter before supervisor review."	Recommendations may be generic or inappropriate.
SOURCE	What evidence should AI use?	"Use the uploaded chapter and rubric as primary evidence."	AI may rely on unsupported general knowledge.
METHOD	What sequence should be followed?	"Extract evidence → compare with rubric → prioritize → recommend."	Complex work may be incomplete or inconsistent.

Layer	Design question	What a strong specification looks like	Common failure if omitted
CONSTRAINTS	What must not happen?	"Do not invent citations or rewrite official CO wording."	Unsupported or unauthorized changes may appear.
OUTPUT	What form should the result take?	"Section Evidence Issue Severity Action."	Useful information may be buried in prose.
EVIDENCE RULE	How should claims be grounded?	"Show supporting section; mark EVIDENCE NOT FOUND."	Confidence can be mistaken for evidence.
VALIDATION	How should the output be challenged?	"Identify contradictions and try to disprove the top finding."	First-pass errors may survive.
HUMAN DECISION	What remains faculty-owned?	"Faculty approves final research interpretation."	AI output may be treated as a final decision.

4. The Faculty Prompt Engineering Template

COPY / ADAPT

ROLE:

Act as [relevant faculty/expert perspective].

GOAL:

Produce [specific deliverable] so that [intended use/decision].

CONTEXT:

This task relates to

[course/research/PhD/assessment/accreditation/administration].

Audience: [who will use the output].

Stage: [where we are in the workflow].

SOURCE:

Use [files/data/policies/rubrics] as the primary evidence.

Authoritative source when conflicts exist: [source].

METHOD:

1. Extract relevant evidence.
2. Analyze/compare/classify against [criteria].
3. Identify gaps, contradictions, risks or patterns.
4. Prioritize the findings.
5. Produce the requested deliverable.
6. Run a validation pass.

CONSTRAINTS:

Do not invent citations, findings, statistics, policies or missing data.
 Do not make decisions that are reserved for faculty/institutional authorities.
 [Add task-specific constraints.]

OUTPUT:

Return [table/report/checklist/blueprint/action plan].
 Required fields: [fields].

EVIDENCE & UNCERTAINTY:

For important findings, identify supporting evidence.
 Label FACT / INFERENCE / RECOMMENDATION.
 State EVIDENCE NOT FOUND when support is unavailable.
 Use confidence only as an indicator for review, not proof.

VALIDATION:

Try to disprove the most important findings.
 Identify contradictions, unsupported assumptions and missing evidence.

HUMAN CHECKPOINT:

End with what I must verify and what decision remains human-owned.

Design principle: the prompt should be only as complex as the task requires. Do not add sections mechanically.

5. Prompt Library — How to Use Each Prompt

Each library entry follows a consistent enterprise-learning pattern: business/academic pain point, objective, input, recommended execution surface, techniques, ready-to-run prompt, expected output and validation. The supplied sample files are synthetic and exist solely to enable independent practice.

5.1 Research Proposal — Identify Defensible Research Gaps

Design element	Specification
Business / academic pain point	Faculty/researchers spend time reading across sources but struggle to distinguish a genuine evidence-based gap from a broad topic.
Required source / input	01_Research_Proposal.docx + 02_Literature_Notes.docx
Recommended execution surface	ChatGPT / approved enterprise AI
Prompting techniques	Role + grounding + decomposition + structured output + adversarial validation
Expected deliverable	Expected: evidence-grounded gap matrix
Validation checkpoint	Expected: evidence-grounded gap matrix. Validate the strongest gap against primary literature before using it in a thesis/proposal.

Ready-to-run prompt

Act as a doctoral research-methodology reviewer.

Using ONLY the supplied research proposal and literature notes:

1. Extract the research problem, objectives, proposed method and expected contribution.
2. Map the major literature themes.
3. Extract limitations explicitly stated in the literature notes.
4. Compare those limitations with the proposed research.
5. Identify candidate research gaps.

Label every gap EXPLICIT or INFERRED.

For every gap, provide the supporting evidence and explain the reasoning.

Output: Gap ID | Theme | Supporting Evidence | Limitation | Candidate Gap | Type | Confidence | Validation Needed.

Do not invent citations, findings, datasets or statistics.

Finally, provide three reasons why the strongest proposed gap could be wrong.

5.2 PhD Thesis Chapter — Examiner Review

Design element	Specification
Business / academic pain point	Faculty supervisors/researchers need a rigorous second-pass review before supervisor, committee or viva review.
Required source / input	03_PhD_Thesis_Chapter.docx + 04_Thesis_Review_Rubric.docx
Recommended execution surface	ChatGPT / approved enterprise AI
Prompting techniques	Role + source grounding + rubric-based evaluation + structured output + adversarial validation
Expected deliverable	Expected: prioritized examiner-style review
Validation checkpoint	Expected: prioritized examiner-style review. Faculty validates every criticism against the original chapter.

Ready-to-run prompt

Act as an external PhD examiner.

Review the uploaded thesis chapter against the uploaded review rubric.

Do not rewrite the chapter initially.

Identify the 10 most consequential issues across problem alignment, evidence, methodology, logic, novelty, limitations and academic writing.

For each issue return:

Section | Evidence | Issue | Severity | Why it matters | Recommended Revision | Validation Question.

Separate FACT from INFERENCE.

Do not invent references, findings or methodological evidence.

Finish with five viva questions that test the weakest areas.

5.3 Literature Review — Convert Summary into Synthesis

Design element	Specification
Business / academic pain point	Literature reviews often become paper-by-paper summaries instead of an analytical synthesis.

Design element	Specification
Required source / input	02_Literature_Notes.docx
Recommended execution surface	ChatGPT / approved enterprise AI
Prompting techniques	Grounding + decomposition + structured output
Expected deliverable	Expected: synthesis matrix
Validation checkpoint	Expected: synthesis matrix. Validate all important findings against the original papers.

Ready-to-run prompt

Transform the supplied literature notes from a paper-by-paper summary into an analytical synthesis.
Cluster studies by theme. For each theme compare objectives, approaches, findings/claims and limitations.
Identify agreements, contradictions and under-explored combinations.
Output: Theme | Sources | Common Finding | Difference/Contradiction | Limitation | Candidate Research Opportunity | Evidence Strength.
Do not create or invent references.

5.4 Journal Manuscript — Reviewer #2

Design element	Specification
Business / academic pain point	Researchers need a critical review before submission but may receive overly supportive AI feedback.
Required source / input	11_Journal_Manuscript.docx
Recommended execution surface	ChatGPT / approved enterprise AI
Prompting techniques	Role + rubric-like criteria + evidence + adversarial critique
Expected deliverable	Expected: reviewer-style critique
Validation checkpoint	Expected: reviewer-style critique. Faculty/researcher decides whether each criticism is valid and how to respond.

Ready-to-run prompt

Act as a rigorous journal reviewer.
Review the manuscript for significance, novelty, methodology, evidence, reproducibility, limitations, overclaiming and clarity.
Separate Major Concerns and Minor Concerns.
For every major concern provide:
Manuscript Evidence | Reviewer Concern | Severity | Why it Matters | Suggested Author Response | Evidence Needed.
Do not invent external citations.
Finish with a publication-readiness checklist.

5.5 Reviewer Comments — Build a Response Strategy

Design element	Specification
Business / academic pain point	Reviewer comments can be long, overlapping and difficult to convert into a revision plan.

Design element	Specification
Required source / input	13_Reviewer_Comments.docx + 11_Journal_Manuscript.docx
Recommended execution surface	ChatGPT / approved enterprise AI
Prompting techniques	Classification + grounding + prioritization + structured output
Expected deliverable	Expected: revision decision matrix
Validation checkpoint	Expected: revision decision matrix. Researcher approves the final response strategy.

Ready-to-run prompt

Using the reviewer comments and manuscript:

1. Group comments by theme.
2. Classify each as clarification, additional analysis, methodological concern, evidence concern or writing issue.
3. Map the comment to relevant manuscript evidence.
4. Recommend revision, additional analysis, explanation or respectful disagreement.
5. Identify evidence/revision needed.

Output: Reviewer Comment | Type | Manuscript Evidence | Recommended Action | Evidence/Revision Needed | Priority.

Do not fabricate analyses that have not been performed.

5.6 Course Design — Audit CO Alignment

Design element	Specification
Business / academic pain point	Faculty may have a syllabus, learning activities and assessments that are not fully aligned.
Required source / input	05_Course_Syllabus_and_COs.docx
Recommended execution surface	ChatGPT / Copilot / approved enterprise AI
Prompting techniques	Context + source grounding + structured mapping + evaluation
Expected deliverable	Expected: CO alignment matrix and improvement priorities
Validation checkpoint	Expected: CO alignment matrix and improvement priorities. Faculty validates curriculum decisions.

Ready-to-run prompt

Act as an experienced engineering curriculum designer.

Audit the uploaded syllabus and course outcomes.

Map each CO to topic coverage, learning activities and assessment evidence.

Identify COs that are over-taught, under-assessed, assessed at an inappropriate cognitive level, or insufficiently evidenced.

Output: CO | Evidence | Topic Coverage | Learning Activity | Assessment Evidence | Cognitive Level | Gap | Recommendation | Faculty Validation.

Do not change official CO wording unless explicitly asked.

5.7 Assessment Design — Build the Blueprint Before Questions

Design element	Specification
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Design element	Specification
Business / academic pain point	Generating questions before checking coverage can produce poor CO alignment, uneven difficulty and inappropriate cognitive levels.
Required source / input	05_Course_Syllabus_and_COs.docx + 06_Assessment_Blueprint.docx
Recommended execution surface	ChatGPT / Word / approved enterprise AI
Prompting techniques	Structured output + constraint + curriculum alignment
Expected deliverable	Expected: reviewable blueprint
Validation checkpoint	Expected: reviewable blueprint. Faculty retains final control over assessment validity and examination decisions.

Ready-to-run prompt

Create an improved assessment blueprint using the supplied syllabus, COs and current blueprint.
 Check CO coverage, topic distribution, cognitive-level balance, difficulty, marks and assessment validity.
 Return: Task ID | CO | Topic | Cognitive Level | Difficulty | Marks | Current Gap | Recommendation | Validation Note.
 Do not generate final examination questions yet.
 End with five faculty checks required before approval.

5.8 Student Feedback — Identify Actionable Patterns

Design element	Specification
Business / academic pain point	Large volumes of feedback consume time and can lead to overreaction to isolated comments.
Required source / input	07_Student_Feedback.xlsx
Recommended execution surface	Excel / ChatGPT / approved enterprise AI
Prompting techniques	Structured classification + frequency analysis + uncertainty
Expected deliverable	Expected: theme/action matrix
Validation checkpoint	Expected: theme/action matrix. Faculty decides whether a pattern is educationally meaningful and what action to take.

Ready-to-run prompt

Analyze the uploaded student feedback.
 Cluster recurring themes and distinguish recurring patterns from isolated comments.
 Do not infer sensitive personal attributes.
 For each theme provide:
 Theme | Frequency | Supporting Comments/IDs | Positive/Concern | Possible Cause (hypothesis) | Recommended Action | Priority | Confidence.
 Clearly label hypotheses as hypotheses.
 Then create a 30-day teaching-improvement action list.

5.9 Accreditation — Evidence Gap Analysis

Design element	Specification
Business / academic pain point	Accreditation preparation can become a document-collection exercise without demonstrating whether evidence actually satisfies the criterion.
Required source / input	08_Accreditation_Criterion_Evidence.docx
Recommended execution surface	ChatGPT / approved enterprise AI
Prompting techniques	Grounding + requirement decomposition + evidence matrix
Expected deliverable	Expected: evidence-gap matrix
Validation checkpoint	Expected: evidence-gap matrix. Institutional/quality team validates the interpretation of the criterion.

Ready-to-run prompt

Act as an academic accreditation evidence reviewer.
 Using ONLY the supplied criterion and evidence:
 1. Extract each requirement.
 2. Map each requirement to available evidence.
 3. Identify missing, weak, ambiguous or disconnected evidence.
 4. Identify where evidence exists but does not demonstrate the requirement.
 Output: Requirement | Evidence | Evidence Location | Strength | Gap |
 Risk/Consequence | Recommended Action | Validation Owner.
 Do not claim compliance merely because a document exists.

5.10 Department Meeting — MOM and Action Tracker

Design element	Specification
Business / academic pain point	Meeting discussions contain suggestions, decisions and actions that are often mixed together in manual MOM preparation.
Required source / input	09_Department_Meeting_Transcript.docx
Recommended execution surface	Teams/Copilot or ChatGPT
Prompting techniques	Extraction + structured output + constraint prompting
Expected deliverable	Expected: reviewable MOM/action register
Validation checkpoint	Expected: reviewable MOM/action register. Meeting owner validates decisions before circulation.

Ready-to-run prompt

Convert the supplied transcript into a governed academic MOM.
 Extract only explicit decisions, commitments and action items.
 Return:
 A. Key discussion points
 B. Decisions explicitly made
 C. Actions: Action | Owner | Due Date | Dependency | Evidence
 D. Open questions
 E. Items requiring HoD/committee attention

Do not convert suggestions into decisions.
If owner or due date is missing, write NOT CONFIRMED.

5.11 Grant Proposal — Reviewer Simulation

Design element	Specification
Business / academic pain point	Grant proposals can be internally polished but still poorly aligned to the funding call.
Required source / input	01_Research_Proposal.docx + 10_Grant_Call.docx
Recommended execution surface	ChatGPT / approved enterprise AI
Prompting techniques	Grounding + role + criteria-based evaluation
Expected deliverable	Expected: proposal readiness matrix
Validation checkpoint	Expected: proposal readiness matrix. Research team verifies all call requirements independently.

Ready-to-run prompt

Act as a research-grant review panel.
Compare the proposal against the supplied grant call.
Evaluate significance, alignment, objectives, methodology, measurable outcomes, feasibility, risks and dissemination.
Output: Criterion | Proposal Evidence | Strength | Concern | Severity | Recommendation | Missing Evidence.
Do not invent eligibility or funding rules beyond the supplied call.
End with five questions most likely to affect competitiveness.

5.12 Faculty Workload — Identify Pressure Points

Design element	Specification
Business / academic pain point	Faculty workload data may reveal concentration and administrative bottlenecks that are difficult to see manually.
Required source / input	12_Faculty_Workload.xlsx
Recommended execution surface	Excel / ChatGPT / approved enterprise AI
Prompting techniques	Data analysis + structured output + decision support
Expected deliverable	Expected: decision-support view, not an automatic staffing decision
Validation checkpoint	Expected: decision-support view, not an automatic staffing decision.

Ready-to-run prompt

Analyze the faculty workload data.
Identify workload concentration, peak constraints, potential single points of failure and areas where administrative work may be streamlined.
Do not automatically recommend changing teaching allocations.
Output: Faculty | Pressure Signal | Evidence | Potential Impact | Workflow Opportunity | Recommendation | Confidence.
End with three questions leadership should answer before taking action.

5.13 Teaching Design — Create Authentic Learning Activities

Design element	Specification
Business / academic pain point	AI can generate activities that are easy to answer but weak in authentic learning value.
Required source / input	05_Course_Syllabus_and_COs.docx
Recommended execution surface	ChatGPT / approved enterprise AI
Prompting techniques	Context + structured output + constraint + responsible AI
Expected deliverable	Expected: activity designs with explicit learning evidence and AI-use boundaries
Validation checkpoint	Expected: activity designs with explicit learning evidence and AI-use boundaries.

Ready-to-run prompt

Using the uploaded syllabus and COs, design three authentic learning activities. For each specify:

CO | Skill | Student Task | Evidence of Learning | Difficulty | Estimated Time | Appropriate AI Use | Misuse Risk | Faculty Validation.

Activities must require student reasoning and should not simply reproduce AI-generated answers.

Include one activity that requires students to critique an AI output.

5.14 Research Workflow — Convert Meeting Input into Controlled Action

Design element	Specification
Business / academic pain point	Research discussions can generate useful ideas that are later mistaken for approved research decisions.
Required source / input	09_Department_Meeting_Transcript.docx + 01_Research_Proposal.docx
Recommended execution surface	ChatGPT / approved enterprise AI
Prompting techniques	Source comparison + classification + constraint + mapping
Expected deliverable	Expected: controlled research-action matrix
Validation checkpoint	Expected: controlled research-action matrix. Researcher/supervisor validates any change to research direction.

Ready-to-run prompt

Compare the department meeting transcript with the research proposal.

Classify each relevant meeting statement as:

- A. Actionable
- B. Question requiring evidence
- C. Suggestion
- D. Decision explicitly made
- E. Unresolved issue

Map each actionable item to the affected proposal section.

Output: Meeting Evidence | Category | Proposal Section | Required Action | Owner |

Validation Needed.

Do not treat discussion or suggestions as approved research direction.

6. How to Engineer a Prompt from Your Pain Point

The most transferable skill is not memorizing prompts. It is diagnosing the work. Start with the pain point, then translate it into an AI task.

Design step	Diagnostic question	Example — student feedback	What the prompt should capture
1. Pain point	Where is time/rework being lost?	Reading and manually grouping 200 comments.	The work bottleneck, not merely the topic.
2. Outcome	What useful artifact should exist afterward?	A prioritized improvement matrix.	A measurable/inspectable deliverable.
3. Inputs	What information already exists?	Feedback spreadsheet.	Files, fields, documents and data.
4. Authority	Which source should win if information conflicts?	The actual feedback data.	Authoritative evidence hierarchy.
5. Task method	What reasoning steps are needed?	Cluster → count → interpret → prioritize.	Explicit workflow stages.
6. Risk	What could go wrong?	AI overreacts to one comment.	Failure modes and constraints.
7. Output	How will I consume the result?	Theme table + 30-day action plan.	Columns, sections, length and audience.
8. Validation	How will I know it is trustworthy enough?	Show comments supporting each theme.	Evidence links, confidence, challenge questions.
9. Human decision	What cannot be delegated?	Whether to change course design.	Decision rights and approval point.

7. Pain Point → Technique Selection

Pain point category	Primary technique	Supporting techniques	Why
Need a quick transformation	Zero-shot	Context + structured output	Fastest route for well-defined tasks.
Need consistent classification	Few-shot	Constraints + structured output	Examples establish the desired pattern.
Need expert critique	Role/perspective	Rubric + evidence + adversarial validation	Makes evaluation criteria explicit.
Need to analyze documents	Grounding/source	Decomposition + structured output	Reduces generic responses and

Pain point category	Primary technique	Supporting techniques	Why
			improves traceability.
Need a research/PhD review	Grounding + role	Decomposition + adversarial validation	Combines evidence with critical evaluation.
Need curriculum alignment	Context + source	Mapping + structured output	Connects COs, activities and assessment.
Need an evidence/compliance review	Grounding	Requirement decomposition + evidence matrix	Separates evidence presence from evidence adequacy.
Need a repeatable workflow	Master prompt	Placeholders + validation	Standardizes a recurring task.

8. Deriving a Master Prompt

A master prompt is a tested, reusable instruction set for a recurring workflow. It should emerge from iteration—not be written as a long prompt on the first attempt.

Iteration	Observed issue	Prompt engineering response	Quality gate
V1 — Baseline	Output is generic.	Add goal, context and role.	Is the output relevant?
V2 — Evidence	Claims are unsupported.	Specify authoritative sources and evidence rules.	Can important findings be traced?
V3 — Structure	Output is difficult to use.	Define a stable schema/table/checklist.	Can the result be reviewed quickly?
V4 — Coverage	Important areas are missed.	Add decomposition/checklist and explicit criteria.	Are all required dimensions covered?
V5 — Robustness	AI agrees too easily.	Add adversarial challenge and contradiction search.	Does the output survive challenge?
V6 — Governance	AI recommendations are treated as decisions.	Add confidence, validation and human checkpoint.	Are decision rights explicit?
V7 — Reuse	Prompt works only for one case.	Replace case-specific details with variables/placeholders.	Can another case use the same prompt?
Master	Workflow is repeatable.	Save prompt + usage instructions + sample input/output + validation checklist.	Can another faculty member execute it consistently?

9. Master Prompt — Faculty Academic Intelligence Assistant

MASTER PROMPT

ROLE

You are my Faculty Academic Intelligence Assistant: a rigorous, evidence-grounded partner for recurring university faculty work.

MISSION

Help me analyze, structure, challenge and improve work across teaching, curriculum, assessment, research, PhD supervision, academic administration and accreditation.

OPERATING PRINCIPLES

1. Start with the requested outcome, not a generic explanation.
2. Use supplied/approved sources as the primary evidence.
3. Distinguish FACT, INFERENCE and RECOMMENDATION.
4. Never fabricate citations, findings, statistics, policies, student information or missing data.
5. State EVIDENCE NOT FOUND when support is unavailable.
6. Use structured outputs when the result will be reviewed or reused.
7. Challenge important findings before finalizing.
8. Keep consequential academic and institutional decisions human-owned.

WORKFLOW

- A. Clarify the task, audience, context and intended use.
- B. Identify the authoritative source(s).
- C. Extract relevant evidence.
- D. Apply the appropriate analysis method: classify, compare, map, synthesize, evaluate, prioritize or decompose.
- E. Identify gaps, contradictions, assumptions and risks.
- F. Produce the requested deliverable in a reviewable structure.
- G. Run an adversarial validation pass.
- H. State what the faculty member must verify and decide.

EVIDENCE RULES

- Do not use confidence as a substitute for evidence.
- For important findings, identify the supporting source/section where available.
- Never convert a suggestion into a decision.
- Never convert an inference into a fact.
- Do not claim external verification unless it actually occurred.

ACADEMIC INTEGRITY

Do not fabricate research data, participant responses, citations or findings.
Do not assist in misrepresenting AI-generated work as independently produced scholarship.

For assessments, preserve student reasoning and faculty oversight.

OUTPUT STANDARD

Prefer tables, matrices, checklists and action plans where appropriate.
Use concise professional academic language.
Include assumptions and evidence gaps explicitly.

ADVERSARIAL VALIDATION

Before finalizing, answer:

1. What could make the conclusion wrong?
2. What evidence contradicts it?
3. Which assumptions are unsupported?
4. What additional evidence is needed?
5. Which parts require human review?

HUMAN CHECKPOINT

End substantial outputs with:

- Faculty verification required:
- Human decision required:
- Evidence to collect next:

If the task is ambiguous or high-risk, ask for clarification rather than inventing assumptions.

10. Master Prompt Builder — Use for Any New Faculty Pain Point

FILL-IN-THE-BLANKS

MY ROLE:

I am a [faculty/research/academic leadership role].

PAIN POINT:

I currently spend too much time/rework on [specific task].

DESIRED OUTCOME:

I need [specific deliverable] to support [specific use/decision].

CONTEXT:

Course/research/project:

Audience:

Stage/deadline:

Relevant constraints:

SOURCE:

I will provide:

Authoritative source:

Source hierarchy if conflicts occur:

METHOD:

- 1.
- 2.
- 3.
- 4.
- 5.

CONSTRAINTS:

Do not:

Do not assume:

Do not invent:

Privacy/academic-integrity requirements:

OUTPUT:

Format:

Required fields:

Length/detail:

Audience/tone:

EVIDENCE:

For important findings, show:

Use FACT / INFERENCE / RECOMMENDATION labels:

How should missing evidence be reported?

VALIDATION:

Try to disprove:

Check for:

Ask me to verify:

HUMAN CHECKPOINT:

What must remain faculty-owned?

FOLLOW-UP:

After the first answer, ask me: [specific improvement question].

11. Prompt Quality Rubric — Self-Assessment

Score your prompt from 0 to 2 for each dimension: 0 = absent, 1 = partly defined, 2 = strong.

A total score is useful for improvement, not a guarantee of output quality.

Dimension	0	1	2	My score
Goal clarity	No clear outcome	Broad outcome	Specific deliverable/use	
Context	None	Some context	Relevant audience/stage/purpose	
Source grounding	No source	Source named but vague	Authoritative source + boundaries	
Method	No method	Some steps	Clear logical workflow	
Constraints	None	General caution	Specific non-negotiable rules	
Output design	Free-form	Partly structured	Stable schema/format	
Evidence rules	None	General request for accuracy	Traceability + missing-evidence rule	
Validation	None	“Check your answer”	Explicit adversarial/validation criteria	
Human checkpoint	None	General review	Decision rights explicitly stated	
Reusability	One-off	Some	Parameterized master	

Dimension	0	1	2	My score
		placeholders	prompt	

12. Independent Practice — Build Your Own Prompt

Choose one recurring task from your own academic role. Complete the canvas, execute it, validate it, and create Version 2.

Prompt design field	Your response
Pain point	
Desired outcome	
Role / perspective	
Context / audience / stage	
Source / evidence	
Method / steps	
Constraints	
Output schema	
Evidence / uncertainty rules	
Validation / adversarial challenge	
Human decision / approval	
Follow-up improvement prompt	

13. Responsible AI and Faculty Governance

Area	Faculty practice
Data protection	Use only information you are authorized to process in the selected AI environment.
Research integrity	Do not fabricate data, citations, participant responses or research findings.
Assessment	Do not delegate final grading or consequential academic decisions to AI.
Citations	Verify important citations and references against the original source.
Evidence	Treat AI-generated research gaps, interpretations and recommendations as items for validation.
Transparency	Follow institutional requirements for disclosure of AI assistance where applicable.
Human accountability	Keep final academic, research, ethics and institutional decisions with authorized people.

14. Quick Reference — Match the Prompt to the Work

Work situation	Recommended starting architecture	Add when needed	Output pattern
Simple summarization	Goal + Source + Output	Context	5–10 bullet summary
Research synthesis	Role + Goal + Source + Method	Decomposition + adversarial validation	Evidence matrix
PhD chapter review	Role + Source + Rubric + Output	Adversarial validation	Examiner review table
Curriculum redesign	Goal + Context + CO source	Mapping + constraints	CO alignment matrix
Assessment design	CO source + rubric + output	Few-shot examples + validation	Assessment blueprint
Student feedback	Source + classification method	Few-shot + frequency + uncertainty	Theme/action matrix
Accreditation	Requirement source + evidence	Requirement decomposition	Evidence-gap matrix
Meeting/MOM	Transcript source + extraction rules	Structured output	Decision/action register
Recurring workflow	Master Prompt Builder	Placeholders + QA checklist	Reusable master prompt

15. Learning Reflection

Complete these after practice:

What faculty task will I stop doing manually or reduce through an AI workflow?

Which prompting technique changed the quality of my result most?

What evidence did I need to supply that I initially assumed AI would know?

What failure did the validation step reveal?

What will I change in my Version 2 prompt?

Where must human judgment remain explicit?

16. Sample File Index

File	Primary use	Prompts that use it
01_Research_Proposal.docx	Research problem/objectives/method	5.1, 5.11, 5.14
02_Literature_Notes.docx	Synthetic literature evidence	5.1, 5.3
03_PhD_Thesis_Chapter.docx	Chapter review	5.2
04_Thesis_Review_Rubric.docx	Review criteria	5.2
05_Course_Syllabus_and_COs.docx	Curriculum/CO evidence	5.6, 5.7, 5.13
06_Assessment_Blueprint.docx	Existing assessment structure	5.7
07_Student_Feedback.xlsx	Feedback data	5.8
08_Accreditation_Criterion_Evidence.docx	Criterion/evidence	5.9
09_Department_Meeting_Transcript.docx	Meeting evidence	5.10, 5.14
10_Grant_Call.docx	Funding-call criteria	5.11
11_Journal_Manuscript.docx	Manuscript review	5.4, 5.5
12_Faculty_Workload.xlsx	Workload data	5.12
13_Reviewer_Comments.docx	Peer-review comments	5.5

17. Final Takeaway

Prompt engineering becomes valuable when it moves beyond clever wording and becomes a repeatable professional workflow. The objective is not to create the longest prompt. The objective is to create the smallest reliable instruction system that gives the AI the right task, the right evidence, the right boundaries, the right output structure and the right human checkpoint.

Pain Point → Evidence → Prompt → Execute → Validate → Refine → Master Prompt

Guide by Gitanjali Bhalerao

18. Execution Pack: Input Files, Prompts and Output Files

Treat every prompt as a mini-workflow. Locate the listed input file(s) in Sample_Files, attach or reference them in your approved AI tool, run the prompt from the matching library entry in Section 5, validate the response, then save the reviewed deliverable using the suggested output filename. Output files are learner-created artifacts, not pre-filled answers. If the AI tool cannot generate a file directly, copy the result into Word or Excel and save it. Use only synthetic practice data unless institutional policy authorizes other data.

Workflow	Input file(s) to attach/reference	Prompt to run	Suggested output filename	Required output structure
5.1 Research gap analysis	01_Research_Proposal.docx; 02_Literature_Notes.docx	Run Section 5.1 prompt	Research_Gap_Analysis_Output.docx	Gap ID; theme; evidence; limitation; candidate gap; explicit/inferred; confidence; validation needed
5.2 Thesis chapter review	03_PhD_Thesis_Chapter.docx; 04_Thesis_Review_Rubric.docx	Run Section 5.2 prompt	PhD_Chapter_Examiner_Review.docx	Section; evidence; issue; severity; why it matters; revision action; validation question; viva questions
5.3 Literature synthesis	02_Literature_Notes.docx	Run Section 5.3 prompt	Literature_Synthesis_Matrix.xlsx	Theme; sources; common finding; contradiction; limitation; research opportunity; evidence strength
5.4 Journal	11_Journal_Manuscript.docx	Run Section	Journal_Reviewer_Critique.docx	Major/minor concern;

manuscript review		on 5.4 prompt		manuscript evidence; severity; impact; suggested action; readiness checklist
5.5 Reviewer response planning	13_Reviewer_Comments.docx; 11_Journal_Manuscript.docx	Run Section 5.5 prompt	Reviewer_Response_Matrix.xlsx	Comment; type; manuscript evidence; recommended action; revision/evidence needed; priority
5.6 Course outcome alignment	05_Course_Syllabus_and_COs.docx	Run Section 5.6 prompt	CO_Alignment_Audit.xlsx	CO; evidence; topic coverage; learning activity; assessment evidence; cognitive level; gap; recommendation
5.7 Assessment blueprint review	05_Course_Syllabus_and_COs.docx; 06_Assessment_Blueprint.docx	Run Section 5.7 prompt	Assessment_Blueprint_Review.xlsx	Task ID; CO; topic; cognitive level; difficulty; marks; gap; recommendation; validation
5.8 Student feedback analysis	07_Student_Feedback.xlsx	Run Section 5.8 prompt	Student_Feedback_Insights.xlsx	Theme; frequency; comment/ID; positive/concern; hypothesis; action; priority; confidence; 30-day

				action plan
5.9 Accreditation evidence review	08_Accreditation_Criterion_Evidence.docx	Run Section 5.9 prompt	Accreditation_Evidence_Gap_Analysis.docx	Requirement; evidence; location; strength; gap; consequence; action; validation owner
5.10 Department meeting MOM	09_Department_Meeting_Transcript.docx	Run Section 5.10 prompt	Department_MOM_and_Action_Register.docx	Discussion; explicit decisions; action; owner; due date; dependency; evidence; open question
5.11 Grant proposal review	01_Research_Proposal.docx; 10_Grant_Call.docx	Run Section 5.11 prompt	Grant_Proposal_Readiness_Review.docx	Criterion; proposal evidence; strength; concern; severity; recommendation; missing evidence
5.12 Faculty workload analysis	12_Faculty_Workload.xlsx	Run Section 5.12 prompt	Faculty_Workload_Insights.xlsx	Faculty; pressure signal; evidence; impact; workflow opportunity; recommendation; confidence
5.13 Authentic learning activities	05_Course_Syllabus_and_COs.docx	Run Section 5.13 prompt	Authentic_Learning_Activities.docx	CO; skill; student task; evidence of learning; difficulty; time;

				appropriate AI use; misuse risk; validation
5.14 Research action tracking	09_Department_Meeting_Transcript.docx; 01_Research_Proposal.docx	Run Section 5.14 prompt	Research_Meeting_Action_Matrix.xlsx	Meeting evidence; category; proposal section; action; owner; validation needed

18.1 Standard execution instructions

1. Open the relevant prompt in Section 5 and copy the complete prompt text.
2. Open your institution-approved AI tool.
3. Attach or reference every input file listed for that workflow.
4. Paste the prompt and execute it. If attachments are unsupported, paste relevant text/tables only when authorized.
5. Validate the response for unsupported claims, missing fields, fabricated citations, contradictions and assumptions.
6. Run this follow-up: "Audit your previous answer against the supplied files. List unsupported claims, missing evidence, contradictions, omitted fields and assumptions presented as facts. Correct the output and mark unresolved items EVIDENCE NOT FOUND."
7. Save the reviewed result using the suggested filename. Use DOCX for narrative reports and XLSX for matrices/action registers.
8. Record reviewer/date and follow institutional rules for disclosure of AI assistance.

18.2 Output file quality checklist

- ☐ Correct input files were used and identified.
- ☐ All required fields/columns are present.
- ☐ Important findings are traceable to source evidence.
- ☐ FACT, INFERENCE and RECOMMENDATION are distinguished.
- ☐ Missing evidence is stated clearly; citations/statistics are not invented.

- ☐ Contradictions and limitations are recorded.
- ☐ A responsible faculty member reviewed the result before academic use.